

Advanced Econometrics – Lab 12 – Mock test

Exercise 1

A young data scientist is an intern at the Ministry of Finance. At the very beginning of her internship, she was asked to propose a simple econometric model for tax revenues (TAX). The intern started with the simple distributed lags model of the form

$$\ln(TAX)_t = \beta_0 + \beta_1 VAT_t + \epsilon_t, \quad (1)$$

where VAT stands for value added tax rate and is not transformed. However, for some reason she ended up with an autoregressive distributed lags model. The data available consisted of seasonally adjusted series.

Call:

```
dynlm(formula = lnTAX ~ L(lnTAX, 1) + VAT, start = 2)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.039074	0.709907	2.85956	0.004	***
L(lnTAX, 1)	0.401341	0.102283	3.96147	7.4e-05	***
VAT	0.100613	0.010669	9.70575	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

- (5%) What could have been an economic (not econometric) reason behind selection of autoregressive lags model instead of simple DL model?
- (5%) What could have been an econometric reason behind selection of autoregressive lags model instead of simple DL model?
- (5%) Referring to the model results, should we keep or drop the lagged taxes from the model? Explain.
- (10%) Are $\ln TAX$ and VAT integrated? Explain.
- (5%) Let's suppose that the system is in the steady state if $VAT^* = 10$. What is the steady state value of $\ln TAX$? Explain.
- (5%) Calculate and interpret the long-term multiplier.
- (5%) Interpret the short-term multiplier.

Tabela 1: Dickey-Fuller test results

ln (TAX)							
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5
0	-3.40	0.01	0.64	0.89	0.75	0.26	0.30
1	-3.77	0.01	0.85	0.98	0.66	0.25	0.24
2	-3.08	0.03	0.92	0.99	0.71	0.18	0.18
3	-3.31	0.02	0.86	0.98	0.99	0.39	0.28

First differences of ln (TAX)							
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5
0	-9.27	0.01	0.93	0.98	0.77	0.27	0.30
1	-6.85	0.01	0.99	0.98	0.82	0.12	0.16
2	-5.01	0.02	0.84	0.96	0.99	0.16	0.19
3	-5.44	0.21	0.84	0.96	0.99	0.94	0.85

KPSS test results for VAT

KPSS Unit Root Test #
#####

KPSS test results for ΔVAT

KPSS Unit Root Test #
#####

Test is of type: mu with 4 lags.

Value of test-statistic is: 0.0597

Critical value for a significance level of:
10pct 5pct 2.5pct 1pct
critical values 0.347 0.463 0.574 0.739

Test is of type: mu with 3 lags.

Value of test-statistic is: 0.0309

Critical value for a significance level of:
10pct 5pct 2.5pct 1pct
critical values 0.347 0.463 0.574 0.739